

Eftekhar Hossain

ASSISTANT PROFESSOR,

DEPARTMENT OF ELECTRONICS AND TELECOMMUNICATION ENGINEERING



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Research Interest

Natural Language Processing, Computer Vision, and Multimodal Deep Learning.

Employment

- 2022 – **Asst. Professor**, Department of Electronics and Telecommunication Engineering (ETE), Chittagong University of Engineering and Technology (CUET), Bangladesh.
- 2019 – 2022 **Lecturer**, Department of Electronics and Telecommunication Engineering (ETE), Chittagong University of Engineering and Technology (CUET), Bangladesh.
- 2018 – 2019 **Lecturer**, Department of Electrical & Electronic Engineering, Premier University, Bangladesh.

Education

- 2014 – 2018 **B.Sc. in Electronics and Telecommunication Engineering**
Chittagong University of Engineering and Technology, Bangladesh.
CGPA: 3.88/4.0, **Merit Position:** 1st
- 2021 – 2023 **MSc. in Electronics and Telecommunication Engineering**
Chittagong University of Engineering and Technology, Bangladesh.
CGPA: 4.0/4.0
Thesis: Investigating Multimodal Features to Identify Multilingual Offense from Social Media Memes

Research Publications

- 1 **Eftekhar Hossain**, Omar Sharif, Mohammed Moshiul Hoque, and Sarah M Preum. “Deciphering Hate: Identifying Hateful Memes and Their Targets”. **PREPRINT. 2024.**
- 2 Shawly Ahsan*, **Eftekhar Hossain***, et al. “A Multimodal Framework to Detect Target Aware Aggression in Memes”. **EACL. 2024.**
- 3 **Eftekhar Hossain**, Omar Sharif, Mohammed Moshiul Hoque, and Sarah M Preum. “Align before Attend: Aligning Visual and Textual Features for Multimodal Hateful Content Detection”. **EACL-SRW. 2024.**
- 4 **Eftekhar Hossain**, Omar Sharif and Mohammed Moshiul Hoque. “MUTE: A Multimodal Dataset for Detecting Hateful Memes”. **AACL-SRW. 2022.**
- 5 **Eftekhar Hossain**, Omar Sharif, Mohammed Moshiul Hoque. “MemoSen: A Multimodal Dataset for Sentiment Analysis of Memes”. *Language Resources and Evaluation Conference (LREC)*. **2022.**
- 6 **Eftekhar Hossain**, Mohammed Moshiul Hoque, Enamul Hoque and Md. Saiful Islam. “A Deep Attentive Multimodal Learning Approach for Disaster Identification from Social Media Posts”. **IEEE ACCESS (Q1, IF-3.367)**. **2022.**

- 7 Omar Sharif, **Eftekhair Hossain**, Mohammed Moshiul Hoque. "M-BAD: A Multilabel Dataset for Detecting Aggressive Texts and Their Targets". *CONSTRAINT Workshop (ACL)*. **2022**.
- 8 **Eftekhair Hossain**, Omar Sharif, Mohammed Moshiul Hoque. "Identification of Multilingual Offense and Troll from Social Media Memes using Weighted Ensemble of Multimodal Features". *Journal of King Saud University-Computer and Information Sciences*. **ELSEVIER (Q1, IF-13.473)**, **2022**.
- 9 **Eftekhair Hossain**, Omar Sharif and Mohammed Moshiul Hoque. "Investigating Visual and Textual Features to Identify Trolls from Multimodal Social Media Memes". *Shared Task Description Paper in Workshop on Speech and Language Technologies for Dravidian Languages (EACL)*. **2021**.
- 10 **Eftekhair Hossain**, Omar Sharif, Mohammed Moshiul Hoque. "Multilingual Code-Mixed Hope Speech Detection using Cross-lingual Representation Learner". *Shared Task Description Paper in Workshop On Language Technology For Equality, Diversity, Inclusion (EACL)*. **2021**.
- 11 Omar Sharif, **Eftekhair Hossain** and Mohammed Moshiul Hoque. "Offensive Language Detection from Multilingual Code-Mixed Text using Transformers". *Shared Task Description Paper in Workshop on Speech and Language Technologies for Dravidian Languages (EACL)*. **2021**.
- 12 Omar Sharif, **Eftekhair Hossain**, Mohammed Moshiul Hoque. "Combating Hostility: Covid-19 Fake News and Hostile Post Detection in Social Media". *Shared Task Description Paper in CONSTRAINT Workshop (AAAI)*. **2021**.
- 13 Omar Sharif, **Eftekhair Hossain** and Mohammed Moshiul Hoque. "TechTexC: Classification of Technical Texts using Convolution and Bidirectional Long Short Term Memory Network". *Proceedings of 17th International Conference on Natural Language Processing (ICON, ACL)*. **2021**.
- 14 **Eftekhair Hossain**, Omar Sharif, Mohammed Moshiul Hoque. "SentiLSTM: A Deep Learning Approach for Sentiment Analysis of Restaurant Reviews". *Proceedings of 20th International Conference on Hybrid Intelligent Systems (HIS , Springer)*. **2020**.

To get full list of publications visit my [Google Scholar](#)

Technical Skills

Languages	Python, C, Matlab, C++
Machine Learning	Scikit-Learn, Tensorflow, Keras, PyTorch
Data Science	Pandas, Seaborn, Numpy, Beautiful Soup
Web Development	Flask, Bootstrap, PHP, MySQL.
Electronics	Proteus, Micro-controller, Mikro C, PicKit.
IDE & Tools	Jupyter, Spyder, $\text{\LaTeX}$, Hugo.

Undergraduate Projects

Final Year Thesis	Approached a System that Can Detect Cancer From Bone MR Images Using Neural Network and Neuro Fuzzy Classifier. Skills Used: MATLAB, Image Processing, Machine Learning.
Web Dev Project	Developed An Online Based Semester Registration System Skills Used: HTML, CSS, SQL, PHP, Xampp
Electronics Project	Designed a Heart Beat Rate Monitoring System Skills Used: Proteus, MikroC, PicKit

Misc

Awards and Achievements

- 2021 **Winner**, Hope speech detection shared task, EACL-2021.
 2nd Runner-up, Multimodal troll meme detection and offensive language classification shared task, EACL-2021
- 2016 **Merit Award**, Obtained Highest CGPA in an Academic Year, CUET.
- 2017 **1st Prize**, Project Show competition, CUET
 Best Web Development Project Award, CUET.

Certification

- June-2020 **Natural Language Processing Nanodegree**, *Udacity*.
- April-2020 **IBM AI Engineering Professional Certificate**, *Coursera*.
- March-2020 **TensorFlow: Data and Deployment Specialization**, *Coursera*.
- Oct-2019 **DeepLearning.AI TensorFlow Developer Specialization**, *Coursera*.
- Sep-2019 **Deep Learning Specialization**, *Coursera*.

Teaching

- 2022 Computer Programming (CSE 181)
 Object Oriented Programming (CSE 284)
- 2021 Neural and Fuzzy Systems (ETE 455)
 Multimedia Communication (ETE 407)
- 2020 Data Structure and Algorithms (CSE 281)
 Information Theory and Coding (ETE 311)
 Antenna and Wave Propagation (ETE 317).
- 2019 Signal and System (EEE 201)
 Electrical Circuits-I (EEE 101).
- 2018 Cellular Mobile Communication (EEE 435).

Invited Training

- June-2022 **AI and Machine Learning with Python**, *CUET IT Bussiness Incubator*.
- Oct-2022 **Machine Learning with Python**, *CUET IT Bussiness Incubator*.

Test Scores

- GRE **Overall**-306 (Q-163, V-143)
- IELTS **Overall**-6.5 (L-7, R-7.5, W-6, S-6)